

Dr. Fabrizio De Santis

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Parma, Italy



Bibliography

Fabrizio De Santis is a **Ph.D. student in Information Technology at the University of Parma**, where he conducts research at the intersection of process science and artificial intelligence. His work focuses on designing and developing **neuro-symbolic methods** that combine the interpretability of symbolic reasoning with the predictive power of machine learning, with applications spanning **predictive process monitoring** and **clinical decision support** in the healthcare domain. De Santis is driven by the goal of building AI systems that are not only accurate, but also transparent, fair, and reliable in real-world settings. His research has touched on problems such as compliance-aware process monitoring, counterfactual fairness in prescriptive analytics, and sepsis prediction from clinical event data. He earned his Bachelor's degree in Computer, Electronic and Communications Engineering and his Master's degree in Computer Engineering, both from the University of Parma. During his Ph.D., he has collaborated with the **Process and Data Science (PADS) group at RWTH Aachen University**, under the supervision of Prof. Dr. Wil van der Aalst, one of the leading figures in the process mining community.

Professional Experience

RWTH Aachen University — PADS Group

December 2024 – May 2025

Visiting Ph.D. Student

Aachen, Germany

- Research visit under the supervision of Prof. Wil van der Aalst
- Worked on neuro-symbolic AI for predictive process monitoring

University of Parma

July 2022 – June 2023

Research Fellow

Parma, Italy

- Applied ML and NLP models to the legal domain
- Developed a multi-label classifier for labeling maxims related to the Italian Penal Code
- Fine-tuned a pre-trained BERT model on a large corpus of legal maxims

Education

University of Parma

November 2023 – Present

Ph.D. in Information Technology

Parma, Italy

- Topic: Artificial Intelligence for Business Processes
- Supervisor: Prof. Francesco Zanichelli

University of Parma

September 2021 – October 2023

M.Sc. in Computer Engineering'

Parma, Italy

- Thesis: Design and development of a recommender system based on NLP and Deep Learning for legal applications
- Supervisor: Prof. Stefano Cagnoni
- Summa Cum Laude

University of Parma

September 2017 – April 2021

B.Sc. in Computer, Electronic and Communications Engineering'

Parma, Italy

- Thesis: A coevolutionary approach to pattern classification based on Genetic Programming
- Supervisor: Prof. Stefano Cagnoni

Publications

Journal Articles

- F. De Santis, G. Park, F. Zanichelli, 2026. **Neuro-symbolic artificial intelligence for real-time sepsis mortality prediction.** *Engineering Applications of Artificial Intelligence*, 2026.

Conference Papers

- F. De Santis, G. Park, W. van der Aalst, F. Zanichelli, 2026. **Compliance-Aware Predictive Process Monitoring: A Neuro-Symbolic Approach.** *In: Advanced Information Systems Engineering - 38th International Conference, CAiSE 2026, Verona, Italy, June 8-12, 2026, Proceedings. In: Lecture Notes in Computer Science, Springer, to appear*
- F. De Santis, G. Park, F. Zanichelli, 2026. **Neuro-Symbolic Learning for Predictive Process Monitoring via Two-Stage Logic Tensor Networks with Rule Pruning.** *Advances in Knowledge Discovery and Data Mining - 30th Pacific-Asia Conference, PAKDD 2026, Hong Kong SAR, China, June 9-12, 2026, Proceedings. In: Lecture Notes in Computer Science, Springer, to appear.*

Skills

Languages : Python, JavaScript, HTML/CSS, SQL, C, C++, Java

Libraries : PyTorch, scikit-learn, Pandas, Numpy

Topics : Machine Learning, Neuro-Symbolic AI, Predictive Process Monitoring, Healthcare AI

Languages

Italian Mother tongue

English B2